ALICE L WALTON SCHOOL OF MEDICINE

Electrocardiogram: Waves, Intervals, and Segments (2 of 2)

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Adapted from: Electrocardiogram Part 2: Waves, Intervals, and Segments

Listen to this Brick

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Learning Objectives (4) Versions

After completing this brick, you will be able to:

1. Explain how normal ECG waves, intervals, and segments work.
2. Explain how to determine heart rate from an ECG tracing.
3. Develop a systematic approach to ECG interpretation.
4. Interpret pa ythm (NSR).

CASE CONNECTION

FS is a 65-year-old male who comes in for a preoperative examination before knee replacement surgery. FS has no concerns to report except his chronic joint pain. His heart rate is 75 beats/min and blood pressure is 126/82 mm Hg. "I can't wait to have this surgery to get some pain relief," he says. "But my surgeon says I need an ECG before she'll do the surgery." You arrange for the ECG to be done and later review the tracing on his computerized record.

What information can you give FS from the ECG? How do you calculate his heart rate from the ECG? Consider your answers as you read, and we'll revisit FS at the end of the brick.

GO TO CONCLUSION

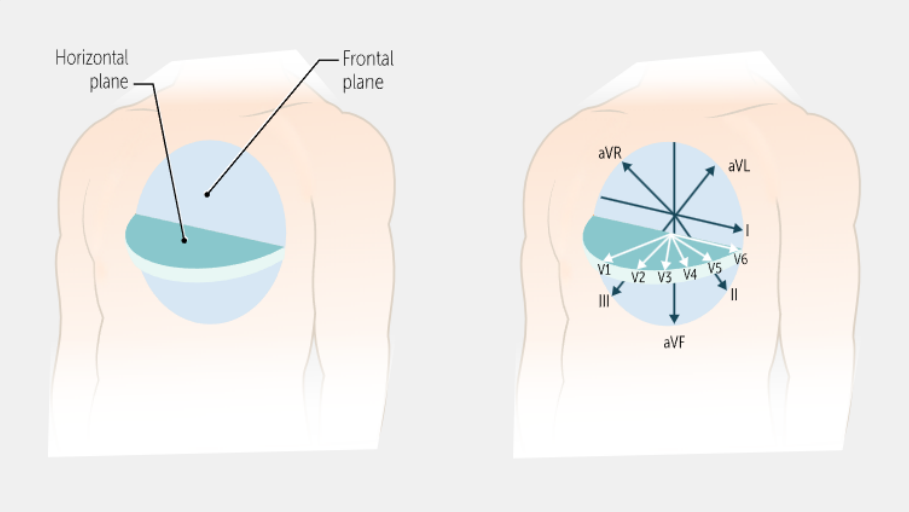
What Is an Electrocardiogram?

Despite the many technological advances in cardiac monitoring during the past 100 years, the electrocardiogram (ECG) remains the cornerstone technique for detecting and monitoring conditions such as myocardial ischemia or infarction, arrhythmias, pericarditis, and drug effects on the myocardium. Once you understand how an ECG works, you can begin to develop a systematic approach to interpreting it.

As Part 2 of the ECG discussion, we focus on interpreting ECG waveforms and segments and how th with some basic techniques for ECG in you start learning about ECG abnormalities.

What Are ECG Waves, Intervals, and Segments? (LO#1)

The normal ECG consists of several waveforms marked by points named in alphabetical order (P through T). For a given ECG lead, an upright (upward pointing) wave means that depolarization is moving toward that electrode, also called a lead (Figure 1).



QUIZ

Figure 1 Heart planes and ECG leads

For most of our discussion, we will use lead II, as it corresponds best to the main vector (direction) of ventricular depolarization. Figure 2 is a real ECG tracing from the perspective of lead II.

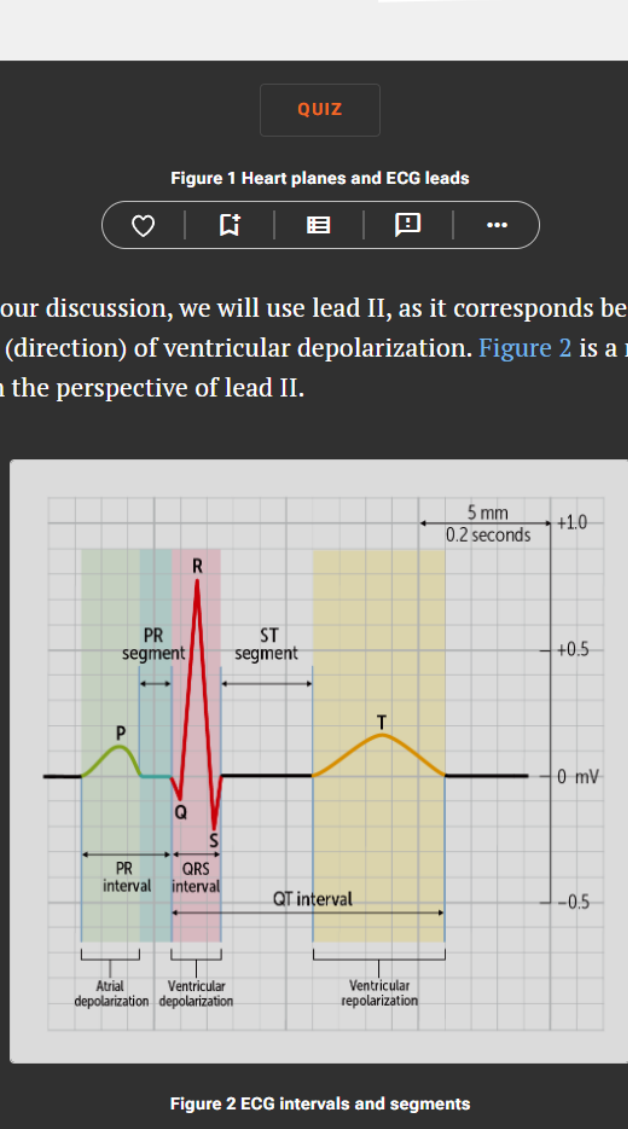


Figure 2 ECG intervals and segments

Waves

The surface electrocardiogram records the electrical activity of the heart as it propagates through t epresents the summed electrical behavior of thousands of cardiac myocytes acting in synchrony and is displayed as characteristic waveforms—the P wave, QRS complex, and T wave—each corresponding to a specific phase of cardiac depolarization or repolarization. Understanding these waves is essential for interpreting normal and abnormal cardiac rhythms.

The **P** wave represents **atrial depolarization** (see Figure 2). During normal conduction, this depolarizing wave front originates from the sinus node which is a cluster of cells located in the upper right atrium along the junction of the superior vena cava and right atrium. Its upward deflection shows that the wave of depolarization is coming toward the lead II electrode (in a southeastern direction, so to speak). The duration of the P wave is usually less than 120 msec (three small boxes). The amplitude of the P wave is <2.5 mm in limb leads and <1.5 mm in precordial leads.

Why don't we see an ECG wave for atrial repolarization? It's there, but it's very small and hidden in the QRS complex.

The following **QRS complex** corresponds to **ventricular depolarization** (see Figure 2). It's much bigger than the P wave, reflecting the much greater mass of myocytes (and volume of electrical signals) in the ventricles than in the atria. It's mostly upward, indicating that the wave of depolarization moving through the ventricles moves largely toward lead II.

The normal duration of the QRS complex is less than 100 milliseconds. The QRS complex is complicated, reflecting the intricacy of the ventricular depolarization pathway. So, it can have anywhere from 1-3 parts depending on which electrode you look at and the individual patient. In Figure 2, on lead II, we see that ventricular depolarization occurs in three phases:

- **Q wave:** the initi lar septum occurs mostly rightward, which is away from lead II, so the ECG deflection is downward (negative).
- **R wave:** the next phase is the depolarization of the mass of the ventricle; the main vector is to the left and inferior, moving directly toward lead II, which is why the deflection is upward (positive) and why it's generally large and dominates the QRS complex.
- **S wave:** finally, depolarization of the posterior surface of the heart goes rightward, which is away from lead II, so the deflection is downward (negative).

The ECG waves and direction of deflection can vary considerably based on the lead and individual patient differences. The QRS normally lasts for 0.06-0.10 seconds.

CLINICAL CORRELATION

Enlarged, "pathologic" Q waves signify underlying scar tissue that occurs after a myocardial infarction. An enlarged R wave often signifies ventricular hypertrophy, which happens when the myocytes enlarge in response to hypertension or valvular disease. Bigger myocytes mean stronger electrical signals and larger ECG waves.

The **T wave** corresponds to **ventricular repolarization**. Why is it normally upward if repolarization is a downward phase of the action potential? If the repolarization moved toward lead II, it would indeed produce a downward deflection as expected. However, it turns out that the repolarization wave moves **away** from lea upward.

CLINICAL CORRELATION

"Repolarization changes" are seen in many cardiac diseases like cardiomyopathy and myocardial infarction. This may cause the T waves to reverse their polarity (eg, downward in lead II) or take on distorted configurations.

Segments and Intervals

On the ECG tracing, a **segment** is a flat (isoelectric) region between two waves. In contrast, an **interval** is a duration of the tracing that includes a segment and at least one wave.

The **PR segment** is measured from the end of the P-wave to the beginning of the QRS complex. It represents the time it takes for the AV node and His-Purkinje system to depolarize.

The **PR interval** includes the P wave plus the PR segment. It starts at the beginning of atrial depolarization and continues until the beginning of ventricular depolarization. Why is there such a delay between the two events? Recall that the impulse does not go directly to the ventricles but instead traverses the **atrioventricular (AV) node**, a specialized structure that slows down the impulse, accounting for the length of the PR interval.

What does the PR segment on an ECG represent?

The PR interval is typically 0.12-0.20 seconds but may increase or decrease with changes in heart rate. This is due to sympathetic and parasympathetic nervous system action on the AV node. The sympathetic system speeds conduction (and shortens the PR interval), whereas the parasympathetic system slows conduction (and lengthens the PR interval). Note that, compared to the lengthy PR interval, the actual depolarization of the ventricle (QRS) happens quickly.

CLINICAL CORRELATION

Of clinical importance, the PR interval is an indicator of the speed of conduction through the atrioventricular (AV) node. First-degree AV block and second-degree AV block type I (Wenckebach) are characterized by prolongation of the PR interval due to delayed conduction within the AV node. In contrast, higher-degree AV block, such as third-degree (complete) heart block, occurs when there is complete failure of r below the His bundle. As a result, pendently, and P waves and QRS complexes demonstrate no fixed relationship.

The **ST segment** is measured from the end of the QRS complex to the beginning of the T wave. It corresponds to the plateau phase of the fast-response cardiac action potential. It is usually isoelectric with a slight upward concavity, like a shallow, upside-down bowl.

CLINICAL CORRELATION

Other ST-segment morphologies, or shapes, may represent myocardial ischemia, myocardial infarction, or pericarditis.

The **QT-interval** is measured from the start of the QRS complex to the end of the T wave (so it also contains the ST segment). It defines the period that includes ventricular depolarization and repolarization. The QT interval is most clinically useful in measuring the rate of ventricular repolarization.

Interpreting the QT interval can be challenging because its duration varies with heart rate. As heart rate increases, cardiac cycles occur more frequently, resulting in compression of the ECG waveforms and shorter measured intervals. Consequently, a normal QT interval at a heart rate of 60 beats per minute is longer than a normal QT interval at 120 beats per minute. To allow meaningful comparison across different heart rates, the QT interval must be adjusted, or normaliz is known as the **corrected QT interval (QTc)**. Several formulas exist to calculate the QTc; the most commonly used is Bazett's formula,

$QTc=QT/\sqrt{RR}$ (QT divided by the square root of RR interval)

QTc= divides the measured QT interval by the square root of the R–R interval.

CLINICAL CORRELATION

A QTc up to 440 ms in men and 460 ms in women is considered normal. QTc values above 500 ms are associated with a substantially increased risk of malignant ventricular arrhythmias in both sexes.

A prolonged QT interval can lead to dangerous arrhythmias such as **torsades de pointes**, characterized by irregular waveforms of alternating amplitudes resembling twisting around a point (Figure 3).

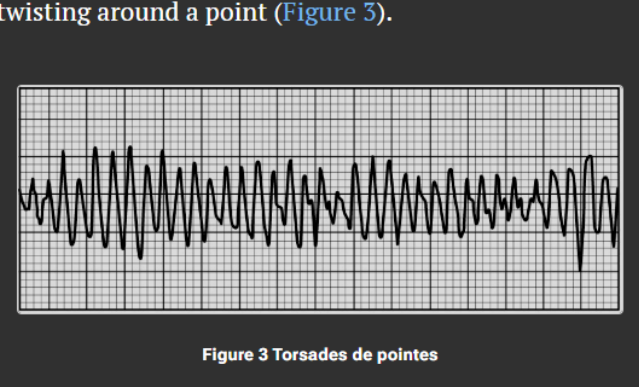


Figure 3 Torsades de pointes

CLINICAL CORRELATION

Some drugs have the side effect of prolonging the QT interval. Sotalol (antiarrhythmic), erythromycin (antibiotic), and haloperidol (antipsychotic) are just a few of the medications that have this side effect. Before these are prescribed, the QT interval should be checked, and the drugs should not be prescribed to someone with a prolonged interval.

What is the danger of a prolonged QT interval?

Figure 4 summarizes the various waves, segments, and intervals.

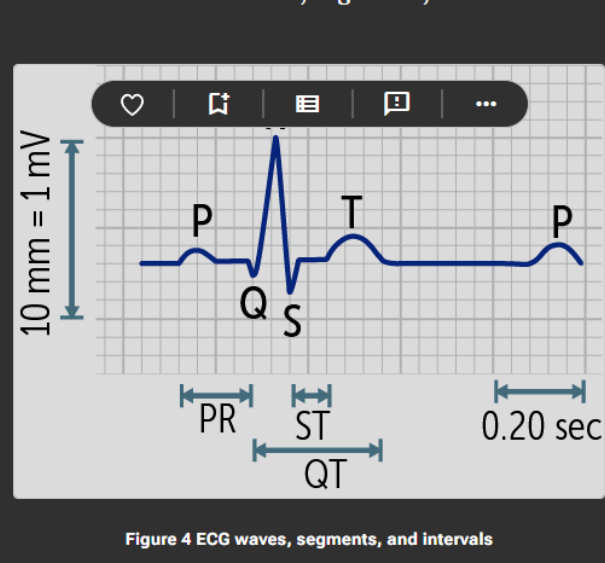


Figure 4 ECG waves, segments, and intervals

How Do ECGs Relate to the Cardiac Cycle?

Now that we've defined and explained our waves and segments, let's go back through them again and correlate each one with an event in the cardiac cycle (Figure 5).

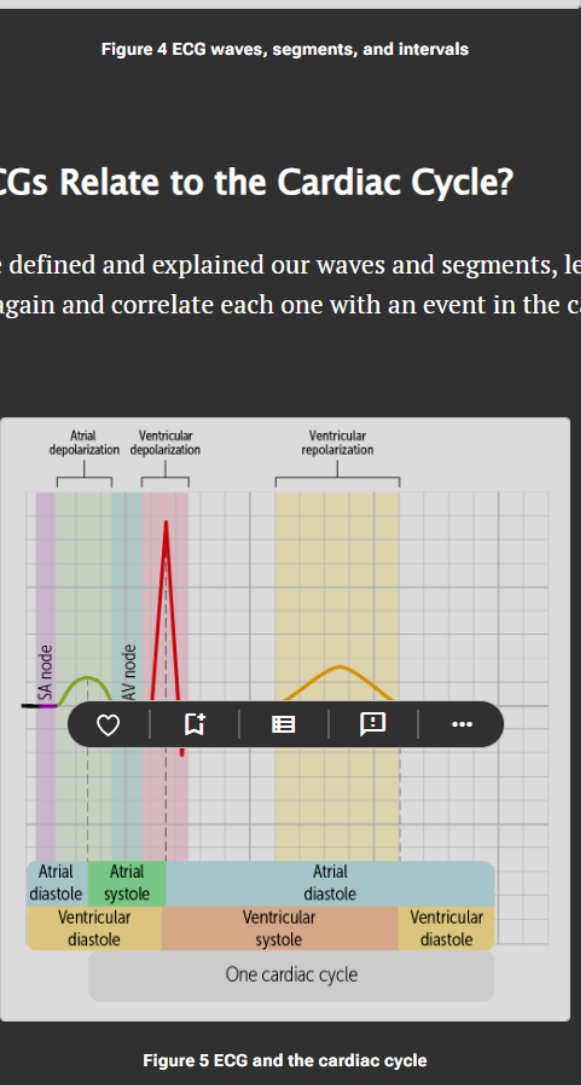


Figure 5 ECG and the cardiac cycle

Note that atrial systole (contraction) starts just after the start of the P wave. The slight delay is expected: the physical act of muscle contraction must await depolarization, calcium entry into myocytes, subsequent calcium release from the sarcoplasmic reticulum, then binding and activation of the actin and myosin filaments.

Atrial systole does not end when the QRS complex is underway (the two events are not directly connected). If the atrial repolarization wave were visible, it would occur just at the start of atrial diastole, in the middle of the QRS complex.

At what point on o start and stop?

For the ventricle, systole begins at about the peak of the R wave and continues all the way through the end of the T wave (ventricular repolarization). So, for both the atrium and the ventricle, the muscle is still contracting even as the electrical repolarization starts, pointing out the delay between the electrical and muscular events.

How Do We Interpret ECGs? (LO#2)

Now that we have covered the ECG grid, including the waveforms, intervals, and segments, we have all the necessary tools to analyze an ECG.

Calculating the Heart Rate

We can calculate a patient's heart rate directly from the ECG. This does not entirely replace taking their pulse. This is because in some diseases, patients show an electrical impulse on ECG but no actual cardiac muscle contraction and therefore no pulse—a condition called electromechanical dissociation or known clinically as pulseless electrical activity (PEA). But usually, the ECG can be used to calculate the heart rate, and this can be done in several ways.

Technique One. One all squares are covered from the beginning of a P wave to the beginning of the next P wave, representing one complete heartbeat (Figure 6).

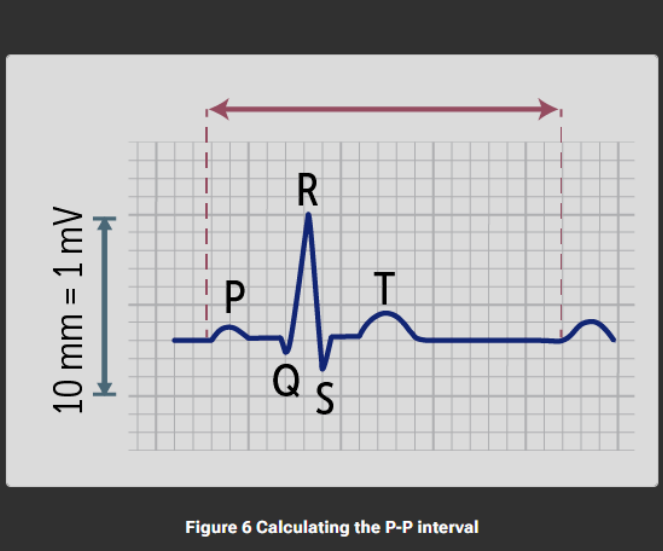


Figure 6 Calculating the P-P interval

Using the number of squares between the two P waves and converting this into seconds, we can calculate the equivalent time value representing the duration of one cardiac cycle. To find out the patient's heart rate or beats per minute, simply divide 60 by the calculated time value.

Let's try an example using the ECG in Figure 7. First count the number of small boxes from the beginning of the first P wave to the beginning of the next P wave: 20. Each small box represents 0.04 seconds, meaning the entire cardiac cycle lasted 0.04 × 20 = 0.8 seconds. To calculate the heart rate or beats per minute (bpm), divide should arrive at a rate of 75 bpm.

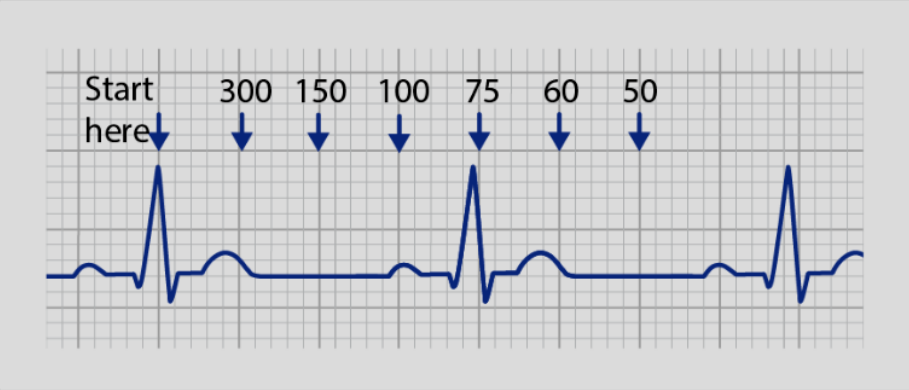


Figure 7 Calculating heart rate

Technique Two. If the previous technique seems tedious, here is a shortcut. Every physician should keep the numbers 300, 150, 100, 75, 60, and 50 in mind. These represent the heart rates of one entire cardiac cycle spanning 1, 2, 3, 4, 5, or 6 large boxes respectively. You can calculate it (rather than memorizing) by dividing 300 by the number of large boxes. Figure 7 shows that the P-to-P interval (or simpler, R-to-R interval) spans four large boxes, representing a heart rate of ~75 bpm. If the R-to-R interval spanned three large boxes, the heart rate would be ~100 bpm, two large boxes would be 150 bpm, and so on.

Keep in mind that in severe conduction disease, such as complete (third-degree) atrioventricular block, atrial and ventricular activity are electrically dissociated and occur at independent rates. In contrast, on a normal ECG with intact 1:1 atrioventri he P wave or the onset of the QRS complex can be used as a consistent reference point for interval measurement.

What is the heart rate if one P-P interval spans five large boxes on an ECG?

Technique Three. This one is the easiest of the three and works when you have a 12-lead ECG, as we saw earlier (Figure 8).

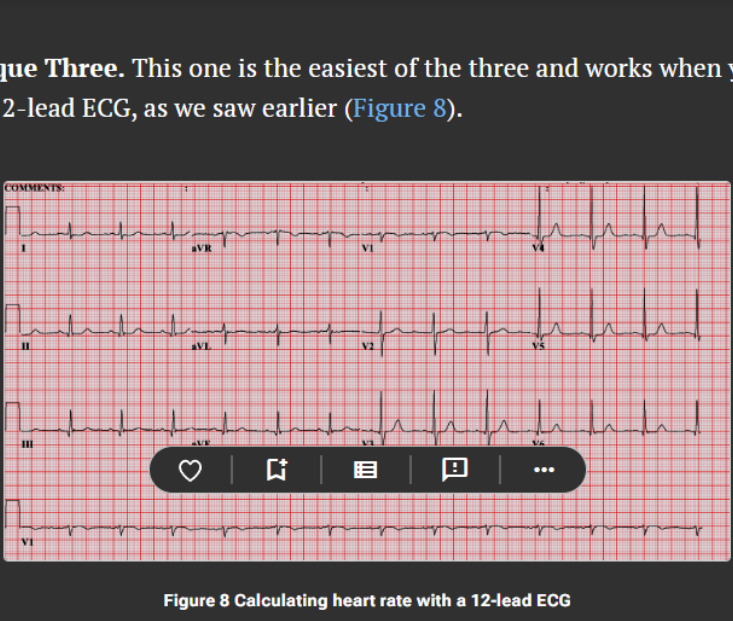


Figure 8 Calculating heart rate with a 12-lead ECG

Since a full tracing is 10 seconds long, simply count the number of beats on the rhythm strip at the bottom and multiply that number by 6. This tracing shows 13 beats, so the rate is 13 x 6 = 78 bpm. Makes sense!

Developing a Systematic Approach to ECG Interpretation (LO#3)

Having a systematic approach to interpreting an ECG is an important part of ensuring we do not miss a critical finding. By reviewing ECGs the same way every time, we build confidence in our skills and increase the speed and accuracy of interpretation. Below is a starting point for a systematic evaluation of a 12 lead ECG. While going through an interpretation scheme, we'll also briefly discuss some cardiac diseases to introduce you to how an ECG can help in diagnosis.

Step 1, Rate: Is the rate 60-100 bpm?

CLINICAL CORRELATION

A heart rate <60 bpm represents bradycardia, whereas a heart rate >100 bpm represents tachycardia.

Step 2, Rhythm: Are the QRS complexes occurring regularly or irregularly? Are there P waves present, and if so, is the morphology consistent with sinus node origin?

CLINICAL CORRELATION

An absent P wave with an irregular rhythm may be seen with atrial fibrillation.

Step 3, Intervals and Segments: Are the PR interval, QRS complex, and QT interval within normal time limits?

CLINICAL CORRELATION

Progressive lengthening or fixed lengthening of the PR interval may be noted with various degrees of heart block.

QRS complex widening may be noted in left or right bundle branch blocks or ventricular preexcitation conditions like Wolff-Parkinson-White syndrome.

QT intervals may be shortened in hypercalcemia and genetic conditions. QT intervals may be lengthened with drugs (antipsychotics, amiodarone, erythromycin), hypokalemia, or certain genetic conditions.

Step 4, Waveforms: Are the shapes of the waveforms the same from one cardiac cycle to the next? Is there any elevation or depression of the ST segment?

CLINICAL CORRELATION

ST-segment elevation may be due to myocardial infarction, whereas a depression may indicate myocardial ischemia (eg, angina).

Peaked T waves may indicate hyperkalemia.

Step 5, Integrate and Interpret: Once you have assessed the ECG with steps 1 through 4, look for an overall diagnosis.

Interpretation of a Normal Sinus Rhythm (LO#4)

Normal Sinus Rhythm (NSR) refers to the heart's electrical activity originating from the sinoatrial (SA) node and progressing normally through the conduction system, producing a regular, coordinated rhythm that is seen on an ECG. Consideration the stepwise approach to analyze an NSR (figure 9)

- *Heart rate* for an adult rate: 60–100 bpm
- Approximate method: 300 ÷ number of large boxes between R waves Rhythm regularity *R–R intervals are regular* Variation should be < 1 small box
- *P wave* present before *every QRS*. Uniform morphology Duration: ≤ 0.12 sec (≤ 3 small boxes) Upright in lead II, inverted in aVR.
- *PR interval* (start 20 sec Equivalent to 3–5 small boxes
- *PR segment* (end of P to start of QRS) Normally isoelectric No specific duration requirement, but should be flat
- *QRS complex* Normal duration: ≤ 0.12 sec Equivalent to ≤ 3 small boxes
- *ST segment* Normally isoelectric
- *T wave* Upright in most leads smooth, asymmetric shape

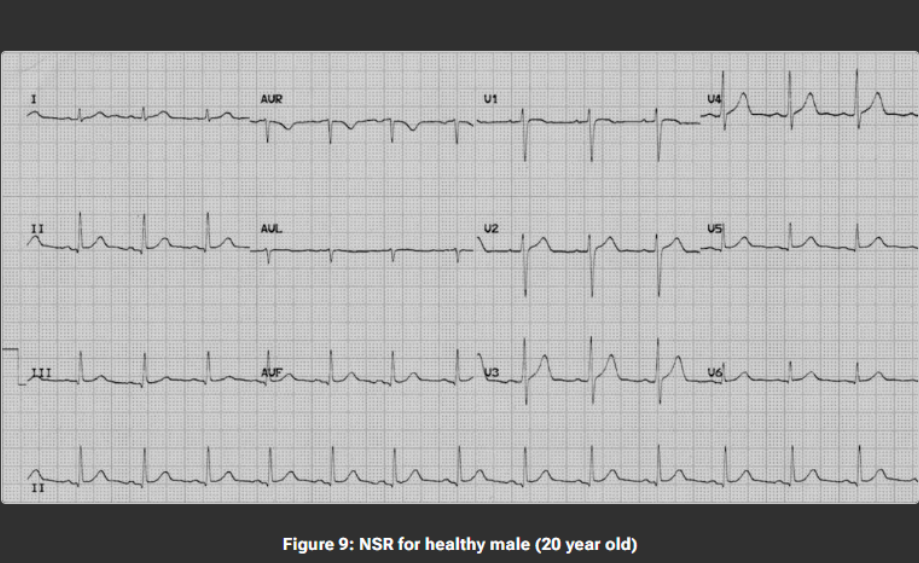


Figure 9: NSR for healthy male (20 year old)

CASE CONNECTION

BACK TO INTRODUCTION

Thinking back to FS, what information do you give him from the ECG? How do you calculate his heart rate?

You systematically review his ECG, determining heart rate, rhythm, intervals, and evidence of prior he e boxes on the tracing, indicating a rate of 75 beats/min—exactly what you found on physical examination. You explain to FS that the ECG has not shown any abnormalities and how this is helpful in assessing the level of risk posed by general anesthesia and major surgery. It can also provide a useful baseline for later comparison if there is any concern about future heart trouble. You call the surgeon to let her know that FS's ECG is normal and did not reveal any contraindication to his knee surgery.

Summary

- An electrocardiogram (ECG) is a key tool for detecting and monitoring cardiac conditions such as myocardial ischemia, arrhythmias, and drug effects on the heart.
- ECG waveforms include the P wave (atrial depolarization), QRS complex (ventricular depolarization), and T wave (ventricular repolarization), each representing specific phases of cardiac activity.
- Segments and intervals on an ECG, such as the PR interval and QT interval, provide insights into the conduction speed through the heart and can indicate conditions like heart block or arrhythmias.
- The ST segment is crucial for identifying myocardial ischemia, infarction, or pericarditis, while the QT interval is important for assessing the risk of ventricular arrhythmias.
- Heart rate can be calculated from an ECG by measuring the time between waveforms or by counting beats on a rhythm strip, with adjustments for heart rate variability.
- A systematic approach to ECG interpretation involves assessing heart rate, rhythm, intervals, segments, and waveforms to identify normal sinus rhythm or poten

Review Questions

1. Why is a Q wave represented as deflecting downward in lead II on an ECG strip?

A. ○ It represents depolarization of the interventricular septum toward the left.
B. ● It represents depolarization of the interventricular septum toward the right.
C. ○ It represents depolarization of the posterior ventricle occurring toward the left.
D. ○ It represents depolarization of the posterior ventricle occurring toward the right.
E. ○ It represents depolarization of the whole muscle bulk of the ventricle.

Hide Explanation

The correct answer is it represents depolarization of the interventricular septum toward the right (B). The Q wave represents initial depolarization of the interventricular septum. It occurs mostly rightward, due to early activation of the septum by the left bundle, so the depolarization wave is inverted (down) due to this wavefront traveling away from lead II. It represents depolarization of the interventricular septum toward the left (A) is incorrect, as the wave usually travels rightwards. It represents depolarization of the posterior ventricle toward the left (C) and it represents depolarization of the posterior ventricle toward the right (D) are incorrect, as depolarization of the posterior ventricle is illustrated by the S wave, which travels rightwa depolarization of the whole muscle bulk of the ventricle (E) is given by the R wave, which has a main vector to the left, and, thus, a positive deflection.

2. During which of the following portions of the ECG does isovolumetric contraction of the left ventricle occur?

A. ○ P wave
B. ○ PR interval
C. ○ PR segment
D. ● QRS complex
E. ○ T wave

Hide Explanation

The correct answer is QRS complex (D). Isovolumetric contraction occurs during the early QRS complex, after ventricular depolarization begins and before ventricular ejection starts. This incorporates the initial part of ventricular systole, as the ventricle contracts against a closed aortic valve (isovolumetric contraction). The P wave (A) preceded atrial contraction, during ventricular diastole. The PR interval (B) is the P wave and PR segment and includes atrial systole (during ventricular diastole). The PR segment (C) extends from the end of the P wave to the start of the QRS complex, and measures conduction through the AV node. Atrial systole occurs here. The T wave (E) ccurs during the latter part of ventricular systole, as blood is ejected from the ventricle.

3. A 65-year-old man undergoes ECG testing as part of a preoperative evaluation. He is asymptomatic and hemodynamically stable. The ECG shows a regular rhythm at 78 beats per minute with a P wave preceding every QRS complex. The PR interval is constant at 0.16 seconds. However, the QRS complexes measure 0.14 seconds. ST segments are isoelectric and T waves are upright. What is abnormal about the ECG?

A. ○ The rhythm is not sinus in origin
B. ○ The PR interval is prolonged
C. ● QRS complex is widened
D. ○ ST segment suggests myocardial ischemia

Hide Explanation

The correct answer is (C) QRS complex is widened. A normal QRS complex duration is ≤ 0.12 seconds, and a QRS duration of 0.14 seconds indicates delayed ventricular depolarization. Choice (A) is incorrect because a P wave before every QRS complex with consistent morphology and a regular rhythm, which supports a sinus origin rather than a non-sinus rhythm. Choice (B) is incorrect because the PR interval is 0.16 seconds, which falls within the normal range of 0.12–0.20 seconds and therefore does not indicate delayed conduction through the atrioventricular node. Choice (D) is incorrect isoelectric, which is normal. ST-segment to suggest myocardial ischemia or infarction.

How would you rate this Brick?

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Check off the objectives when you feel comfortable with the concept
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Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Explain how normal ECG waves, intervals, and segments work.~~
02 ☑ ~~Explain how to determine heart rate from an ECG tracing.~~
03 ☑ ~~Develop a systematic approach to ECG interpretation.~~
04 ☑ ~~Interpret patterns associated with a normal sinus rhythm (NSR).~~

Objective Confidence: 4/4